

Power Systems Flexibility from District Heating Networks

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Abstract—Coupling heat and electricity systems can improve the overall flexibility of the energy system and facilitate the large-scale penetration of renewable energy sources. In order to optimally exploit the operational flexibility of District Heating Networks (DHN), we introduce a novel mixed integer nonlinear formulation of the combined heat and power dispatch, accounting for pipeline energy storage. To cope with the non-convexities and computational complexity of this model we propose mixed integer convex quadratic and mixed integer linear relaxations. The proposed model is implemented on a six-bus electricity and three-node district heating network and compared to a conventional economic dispatch model, in order to evaluate the impact of the DHN on the power system.

Index Terms—integrated energy systems, power system flexibility, convex relaxation, mixed integer second order cone programming, mixed integer linear programming

NOMENCLATURE

Sets and Indexes

$\mathcal{T}$	Set of time periods
$\mathcal{I}^{HS}$	Set of heat stations
$\mathcal{I}^{HES}$	Set of heat exchanger stations
$\mathcal{I}^{CHP}$	Set of combined heat and power plants
$\mathcal{I}^{HP}$	Set of heat pumps $\mathcal{I}^{HP} \subset \mathcal{I}^{HS}$
$\mathcal{I}^E$	Set of electricity generators
$\mathcal{I}^N$	Set of nodes in the district heating network
$\tilde{\mathcal{I}}^{N-}$	Set of nodes with a single pipe arriving $\tilde{\mathcal{I}}^{N-} \subset \mathcal{I}^N$
$\tilde{\mathcal{I}}^{N+}$	Set of nodes with a single pipe departing $\tilde{\mathcal{I}}^{N+} \subset \mathcal{I}^N$
$\mathcal{I}^B$	Set of electricity buses
$\mathcal{I}^P$	Set of heat pipelines
S_n^{P+}/S_n^{P-}	Set of pipes starting/ending at node n
S_n^{HS}	Set of heat stations connected to node n
S_n^{HES}	Set of heat exchanger stations connected to node n
S_n^{HP}	Set of heat pumps connected to node n
S_n^E	Set of electricity generators connected to bus n
S_n^B	Set of electricity buses connected to bus n

Input Parameters

R_p/L_p	Radius/Length of pipe p (m)
μ_p	Thermal loss coefficient in pipe p ($\text{J.m}^{-2}.\text{s}^{-2}.\text{K}^{-1}$)
ρ	Density of water (kg.m^{-3})
ν_p	Pressure loss coefficient in pipeline p
c	Specific heat capacity of water ($\text{J.kg}^{-1}.\text{K}^{-1}$)
η_j^{pump}	Efficiency of water pump in heat station j
Δt	Time intervals (s)
$\underline{mf}_{it}^{HES}/\overline{mf}_{it}^{HES}$	Minimum/maximum mass flow rates at heat exchanger station i (kg/s)
$\underline{mf}_{jt}^{HS}/\overline{mf}_{jt}^{HS}$	Minimum/maximum mass flow rates at heat station j (kg/s)
$\underline{mf}_{pt}^S/\overline{mf}_{pt}^S$	Minimum/maximum mass flow rates in the supply network (kg/s)
$\underline{mf}_{pt}^R/\overline{mf}_{pt}^R$	Minimum/maximum mass flow rates in the return network (kg/s)
$\underline{T}_{nt}^S/\overline{T}_{nt}^S$	Supply temperature bounds at node n (K)
$\underline{T}_{nt}^R/\overline{T}_{nt}^R$	Return temperature bounds at node n (K)
$\underline{pr}_{nt}^S/\overline{pr}_{nt}^S$	Pressure bounds at node n in the supply network (Pa)
$\underline{pr}_{nt}^R/\overline{pr}_{nt}^R$	Pressure bounds at node n in the return network (Pa)
$\underline{pr}_i^{HES}$	Minimum pressure difference at heat exchanger station i (Pa)
$\overline{Q}_{jt}$	Maximum heat output of heat station j (W)
α_j	Marginal cost parameter of heat station or electricity producer j (\$/Wh)
ρ_j^H/ρ_j^E	Heat/electricity fuel efficiency of CHP j
r_j^0/r_j	Heat and electricity outputs ratio of CHP j
$\overline{F}_j$	Maximum fuel consumption of CHP j (W)
L_{nt}^E	Electricity load at bus n (W)
L_{it}^H	Heat load at heat exchanger station i (W)
B_{nm}	Susceptance of line connecting buses n

$\bar{P}_{jt}$	Maximum power output of electricity generator j (W)
$\bar{f}_{nm}$	Maximum flow in line connecting buses n and m (W)
COP_{jt}	Coefficient of performance of heat pump j

Decision variables

$T_{pt}^{S,in}/T_{pt}^{S,out}$	Inlet/outlet temperatures in the supply network (K)
$T_{pt}^{R,in}/T_{pt}^{R,out}$	Inlet/outlet temperatures in the return network (K)
T_{nt}^S/T_{nt}^R	Supply/return temperatures at node n (K)
m_{pt}^S/m_{pt}^R	Mass flow rate in the supply/return network (kg.m ⁻³)
m_{jt}^{HS}	Mass flow rate at heat station j (kg.m ⁻³)
m_{it}^{HES}	Mass flow rate at heat exchanger station i (kg.m ⁻³)
pr_{nt}^S/pr_{nt}^R	Pressure at node n in the supply/return network (Pa)
τ_{pt}^S/τ_{pt}^R	Time delay in the supply/return network (h)
Q_{jt}	Heat production of heat station j (Wh)
P_{jt}	Electricity production of electricity generator or CHP j (Wh)
L_{jt}^{HP}	Electricity consumption of heat pump j (Wh)
L_{jt}^{pump}	Electricity consumption of water pump in heat station j (Wh)
θ_{nt}	Voltage angle at bus n (rad)

I. INTRODUCTION

The rapid growth of renewable production has increased the need for flexibility in power systems. This issue can be partially addressed by coupling different energy systems, such as natural gas, transportation, and district heating [1]. In particular, in systems with a high share of Combined Heat and Power (CHP) plants and Heat Pumps (HP), harnessing the flexibility from the District Heating (DH) system can play a major role in renewable integration [2], [3].

Heat and electricity systems have traditionally been operated sequentially. Due to the strong linkage between heat and electricity outputs, the flexibility of CHPs and HPs is limited when participating in electricity markets [4]–[6]. This lack of flexibility in the power system can result in a decrease of wind power utilization [7]. The authors in [8] showed that a Combined Heat and Power Dispatch (CHDP) can increase the flexibility of CHPs and facilitate the large-scale penetration of renewable energy sources. Additionally, Refs. [9] and [10] introduced a CHPD model accounting for time delays of temperature changes in District Heating Networks (DHNs). These papers showed that the energy storage capacity of DHNs can provide operational flexibility to the overall energy system. However, modeling complex dynamics in DHNs

remains a major challenge. Assuming constant mass flow rates, the authors in [9] introduced a Mixed Integer Linear Programming (MILP) formulation of the CHPD. However, in practice most DHNs are operated using varying mass flows to meet the demand. The CHPD model proposed in [10] exploits the energy storage capacity of DHNs by allowing varying mass flow rates and time delays. Due to the non-convexities introduced by the time delays, the resulting Mixed Integer Nonlinear Problem (MINLP) is computationally intensive and challenging to solve using commercial solvers. The authors in [10] introduced a Benders decomposition algorithm to solve the CHPD. However, for non-convex problems, this approach does not guarantee convergence to a global optimum.

An alternative approach to efficiently solve non convex optimization problems is convex relaxation. While the Benders decomposition approach introduced in [10] is a heuristic method, iteratively converging to a local optimum of the original MINLP, our approach provides a global optimum of a convex relaxation of the original problem. Ref. [11] and [12] summarize recent advances in convex relaxation of optimal power flow problems. Two approaches are introduced, Semidefinite Programming and Second-Order Cone Programming (SOCP) relaxations. The authors in [13] introduced a set of new convex quadratic relaxations for MINLPs applied to power systems. Furthermore, the authors in [14] proposed a Mixed Integer Second-Order Cone Programming (MISOCP) relaxation of a natural gas expansion problem, based on McCormick relaxations for bilinear functions [15]. To date, convex relaxation has not been applied to DHNs.

In view of the state of the art, the contributions of this paper are threefold:

- We first introduce a novel formulation of the CHPD problem, accounting for the energy storage capacity of pipelines. In order to optimally exploit the flexibility from DHNs we consider both temperatures and mass flow rates as decision variables. Contrary to the CHPD described in [10], we introduce a Taylor series expansion of the solution of the heat propagation equation, resulting in a bilinear mixed integer optimization problem. This approach is consistent with small time delays and reduces the computational complexity of the optimization problem.
- To cope with the non-convexities of the proposed model, we introduce MISOCP and MILP relaxations of the original bilinear mixed integer optimization problem.
- We validate our approach by comparing it with a conventional economic dispatch on a small case study.

This paper is organized as follows. We introduce the novel CHPD model in Section II and the two relaxations in Section III. In Section IV, we implement the proposed CHPD model on a simple case study. We conclude in Section V.

II. COMBINED HEAT AND POWER DISPATCH

A. District Heating Network

We consider a double-pipe network with supply and return pipelines. Figure 1 shows a section of the DHN. In practice, the

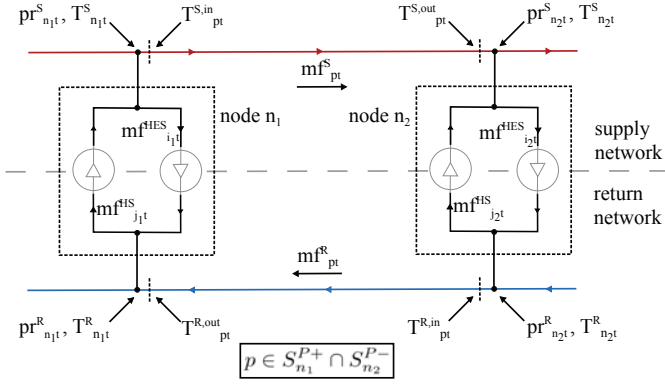


Figure 1. General structure of nodes and double-pipe DHN

control strategies in most DHNs adjust mass flow rates to meet heat demand, while keeping supply and return temperatures within desired levels. These bounds are adjusted depending on ambient temperatures to limit thermal losses and reduced heat demands. In order to optimally exploit the energy storage capacity of DHNs, we consider both temperatures and mass flow rates as decision variables.

Mass flow rates in supply and return networks are constrained by

$$\underline{mf}_{pt}^S \leq mf_{pt}^S \leq \overline{mf}_{pt}^S, \quad \underline{mf}_{pt}^R \leq mf_{pt}^R \leq \overline{mf}_{pt}^R \quad \forall p \in \mathcal{I}^P, t \in \mathcal{T}. \quad (1)$$

Pressure loss due to friction along the pipes can be expressed as

$$pr_{n1t}^S - pr_{n2t}^S = \nu_p (mf_{pt}^S)^2, \quad pr_{n2t}^R - pr_{n1t}^R = \nu_p (mf_{pt}^R)^2 \\ \forall n_1, n_2 \in \mathcal{I}^N, p \in S_{n1}^{P+} \cap S_{n2}^{P-}, t \in \mathcal{T}. \quad (2)$$

The inlet temperatures of the pipes are defined as

$$T_{pt}^{S,in} = T_{nt}^S \quad \forall n \in \mathcal{I}^N, p \in S_n^{P+}, t \in \mathcal{T} \quad (3)$$

$$T_{pt}^{R,in} = T_{nt}^R \quad \forall n \in \mathcal{I}^N, p \in S_n^{P-}, t \in \mathcal{T}. \quad (4)$$

The main challenge is to model the temperature dynamics in the pipelines, and express the outlet temperatures and time delays. Ref. [16] provides an exact solution of the heat propagation equation, which can be approximated using a first order Taylor series expansion:

$$T_{pt}^{S,out} = T_{p(t-\tau_{pt}^S)}^{S,in} \left(1 - \frac{2\mu_p}{c\rho R_p} \tau_{pt}^S \right) \quad \forall p \in \mathcal{I}^P, t \in \mathcal{T}. \quad (5)$$

The discrete time delays τ_{pt}^S are defined as

$$\tau_{pt}^S = \min \left\{ \phi \in \{0, \dots, \bar{\tau}_p^S\}, \text{ s.t. } \sum_{k=t-\phi}^t \frac{mf_{pk}^S}{\pi R_p^2 \rho} \Delta t \geq L_p \right\} \\ \forall p \in \mathcal{I}^P, t \in \mathcal{T}, \quad (6)$$

and the maximum time delays $\bar{\tau}_p^S$ can be computed using the pipelines' physical characteristics.

Eq. (6) represents the time that it takes for the hot water to travel along the pipelines. In order to reformulate this

constraint linearly, we introduce the auxiliary binary variables $u_{p\phi t}^S$, and the constraints

$$M(u_{p\phi t}^S - 1) \leq \sum_{k=t-\phi}^t \frac{mf_{pk}^S}{\pi R_p^2 \rho} \Delta t - L_p < M u_{p\phi t}^S \quad \forall p \in \mathcal{I}^P, \\ \phi \in \{0, \dots, \bar{\tau}_p^S\}, t \in \mathcal{T}, \quad (7)$$

where M represents a big-enough constant. As a result, time delays can be expressed as

$$\tau_{pt}^S = \sum_{\phi=1}^{\bar{\tau}_p^S} \phi (u_{p\phi t}^S - u_{p(\phi-1)t}^S) \quad \forall p \in \mathcal{I}^P, t \in \mathcal{T}. \quad (8)$$

Eq. 5 expresses the outlet temperature of the pipelines, in function of the inlet temperature at time $t - \tau_{pt}^S$. This constraint is non convex due to the bilinear terms $T_{p(t-\tau_{pt}^S)}^{S,in} \tau_{pt}^S$ and the presence of the varying time delays τ_{pt}^S as time indexes. In order to linearize this equation, we introduce the variables $\tilde{T}_{p\phi t}^{S,in}$ and the binary variables $v_{p\phi t}^S$, such that

$$-M v_{p\phi t}^S \leq \tilde{T}_{p\phi t}^{S,in} \leq M v_{p\phi t}^S \quad \forall p \in \mathcal{I}^P, \phi \in \{0, \dots, \bar{\tau}_p^S\}, \\ t \in \mathcal{T} \quad (9)$$

$$M(v_{p\phi t}^S - 1) \leq \tilde{T}_{p\phi t}^{S,in} - T_{p(t-\phi)}^{S,in} \left(1 - \frac{2\mu_p}{c\rho R_p} \phi \right) \\ \leq M(1 - v_{p\phi t}^S) \quad \forall p \in \mathcal{I}^P, \phi \in \{0, \dots, \bar{\tau}_p^S\}, t \in \mathcal{T} \quad (10)$$

$$M(v_{p\phi t}^S - 1) \leq \phi - \tau_{pt}^S \leq M(1 - v_{p\phi t}^S) \quad \forall p \in \mathcal{I}^P, \\ \phi \in \{0, \dots, \bar{\tau}_p^S\}, t \in \mathcal{T} \quad (11)$$

$$\sum_{\phi=0}^{\bar{\tau}_p^S} v_{p\phi t}^S = 1 \quad \forall p \in \mathcal{I}^P, t \in \mathcal{T}. \quad (12)$$

It results from (9)-(12) that $v_{pnt}^S = 1$ and $\tilde{T}_{p\phi t}^{S,in} \neq 0$ if and only if $\tau_{pt}^S = \phi$. In that case, $\tilde{T}_{p\phi t}^{S,in} = T_{p(t-\phi)}^{S,in} \left(1 - \frac{2\mu_p}{c\rho R_p} \phi \right)$. An equivalent linear formulation of Eq. 5 is given by

$$T_{pt}^{S,out} = \sum_{\phi=0}^{\bar{\tau}_p^S} \tilde{T}_{p\phi t}^{S,in} \quad \forall p \in \mathcal{I}^P, t \in \mathcal{T}. \quad (13)$$

We can similarly model temperature dynamics and time delays in return pipelines. However, we omit this step for concision.

B. Nodal Constraints

For incompressible water flows, net mass flow rates at each node are equal to zero, such that

$$\sum_{p \in S_n^{P-}} mf_{pt}^S + \sum_{j \in S_n^{HS}} mf_{jt}^{HS} = \sum_{p \in S_n^{P+}} mf_{pt}^S + mf_{nt}^{HES}, \\ \sum_{p \in S_n^{P-}} mf_{pt}^R + \sum_{j \in S_n^{HS}} mf_{jt}^{HS} = \sum_{p \in S_n^{P+}} mf_{pt}^R + mf_{nt}^{HES} \\ \forall n \in \mathcal{I}^N, t \in \mathcal{T}. \quad (14)$$

Supply and return pressures at each node are bounded by

$$\underline{pr}_{nt}^S \leq pr_{nt}^S \leq \overline{pr}_{nt}^S, \quad \underline{pr}_{nt}^R \leq pr_{nt}^R \leq \overline{pr}_{nt}^R \quad \forall n \in \mathcal{I}^N, t \in \mathcal{T}. \quad (15)$$

Supply and return temperatures at a node are a mix of the outlet temperatures of the pipes arriving at this node

$$T_{nt}^S \sum_{p \in S_n^{P-}} mf_{pt}^S = \sum_{p \in S_n^{P-}} mf_{pt}^S T_{pt}^{S,out} \quad \forall n \in \mathcal{I}^N, t \in \mathcal{T} \quad (16)$$

$$T_{nt}^R \sum_{p \in S_n^{P+}} mf_{pt}^R = \sum_{p \in S_n^{P+}} mf_{pt}^R T_{pt}^{R,out} \quad \forall n \in \mathcal{I}^N, t \in \mathcal{T}, \quad (17)$$

and should be kept within desired bounds

$$\underline{T}_{nt}^S \leq T_{nt}^S \leq \bar{T}_{nt}^S, \quad \underline{T}_{nt}^R \leq T_{nt}^R \leq \bar{T}_{nt}^R \quad \forall n \in \mathcal{I}^N, t \in \mathcal{T}. \quad (18)$$

Observe that for nodes with a single pipe arriving or departing, Equations (16) and (17) can be replaced by the linear constraints

$$T_{nt}^S = T_{pt}^{S,out} \quad \forall n \in \tilde{\mathcal{I}}^{N-}, p \in S_n^{P-}, t \in \mathcal{T} \quad (19)$$

$$T_{nt}^R = T_{pt}^{R,out} \quad \forall n \in \tilde{\mathcal{I}}^{N+}, p \in S_n^{P+}, t \in \mathcal{T}. \quad (20)$$

C. Heat Exchanger Stations

Heat Exchanger Stations (HES) are modeled as heat loads

$$L_{it}^H = cmf_{it}^{HES} (T_{nt}^S - T_{nt}^R) \quad \forall n \in \mathcal{I}^N, i \in S_n^{HES}, t \in \mathcal{T}. \quad (21)$$

The mass flows rates are bounded by

$$\underline{mf}_{it}^{HES} \leq mf_{it}^{HES} \leq \overline{mf}_{it}^{HES} \quad \forall i \in \mathcal{I}^{HES}, t \in \mathcal{T}. \quad (22)$$

and pressure differences between supply and return networks at HESs have to be greater than a threshold in order to ensure that mass flows:

$$pr_{nt}^S - pr_{nt}^R \geq \underline{pr}_i^{HES} \quad \forall n \in \mathcal{I}^N, i \in S_n^{HES}, t \in \mathcal{T}. \quad (23)$$

D. Heat Stations

Heat Stations (HS) represent heat production units, such as CHPs, heat-only units, and HPs. The heat production of a HS is defined as

$$Q_{jt} = cmf_{jt}^{HS} (T_{nt}^S - T_{nt}^R) \quad j \in \mathcal{I}^{HS}, t \in \mathcal{T}, \quad (24)$$

and bounded by

$$0 \leq Q_{jt} \leq \bar{Q}_{jt} \quad \forall j \in \mathcal{I}^{HS}, t \in \mathcal{T}. \quad (25)$$

The mass flow rates are also bounded by

$$\underline{mf}_{jt}^{HS} \leq mf_{jt}^{HS} \leq \overline{mf}_{jt}^{HS} \quad \forall n \in \mathcal{I}^N, j \in S_n^{HS}, t \in \mathcal{T}. \quad (26)$$

The power consumption of a water pump located at HS j is proportional to the pressure difference between supply and return networks, such that

$$L_{jt}^{pump} = \frac{mf_{jt}^{HS} (pr_{nt}^S - pr_{nt}^R)}{\rho \eta_j^{pump}} \quad \forall n \in \mathcal{I}^N, j \in S_n^{HS}, t \in \mathcal{T}. \quad (27)$$

The joint feasible operating region of an extraction CHP links its heat and electricity outputs, such that

$$P_{jt} \geq r_j Q_{jt} \quad \forall j \in \mathcal{I}^{CHP}, t \in \mathcal{T} \quad (28)$$

$$0 \leq \rho_j^E P_{jt} + \rho_j^H Q_{jt} \leq \bar{F}_j \quad \forall j \in \mathcal{I}^{CHP}, t \in \mathcal{T}. \quad (29)$$

The power consumption and heat production of HPs are related by the Coefficient Of Performance (COP), such that

$$Q_{jt} = COP_{jt} L_{jt}^{HP} \quad \forall j \in \mathcal{I}^{HP}, t \in \mathcal{T}. \quad (30)$$

The COP depends on supply and return temperatures. However, it can be assumed to be a known parameter at each time step, in which case Eq. (30) is linear.

E. Power System

As the focus of this study is to model the flexibility provided by DHNs, we model the power transmission network using the standard linearized power flow. The power balance equations at each bus n are

$$\begin{aligned} L_{nt}^E + \sum_{j \in S_n^{HP}} L_{jt}^{HP} + \sum_{j \in S_n^{HS}} L_{jt}^{pump} \\ = \sum_{j \in S_n^E} P_{jt} + \sum_{m \in S_n^B} B_{nm} (\theta_{mt} - \theta_{nt}) \quad \forall n \in \mathcal{I}^B, t \in \mathcal{T}, \end{aligned} \quad (31)$$

where power flows between two nodes are bounded by

$$-\bar{f}_{nm} \leq B_{nm} (\theta_{mt} - \theta_{nt}) \leq \bar{f}_{nm} \quad \forall n \in \mathcal{I}^B, m \in S_n^B, t \in \mathcal{T}, \quad (32)$$

and the power outputs of electricity generators are bounded by

$$0 \leq P_{jt} \leq \bar{P}_{jt} \quad \forall j \in \mathcal{I}^E, t \in \mathcal{T}. \quad (33)$$

This approach can be generalized in order to accommodate more detailed models of electricity transmission networks, including convex relaxations.

F. Production Costs

We approximate the production cost of CHPs as a linear function of the fuel consumption, such that

$$Cost(P_{jt}, Q_{jt}) = \alpha_j (\rho_j^E P_{jt} + \rho_j^H Q_{jt}) \quad \forall j \in \mathcal{I}^{CHP}, t \in \mathcal{T}. \quad (34)$$

Similarly, production costs of heat-only units and electricity generators are assumed to be linear.

As a result, the objective of the CHPD is to minimize the production cost of the overall energy system:

$$\begin{aligned} \sum_{t \in \mathcal{T}} \left(\sum_{j \in \mathcal{I}^E} \alpha_j P_{jt} + \sum_{j \in \mathcal{I}^{HS} \setminus \{\mathcal{I}^{CHP} \cup \mathcal{I}^{HP}\}} \alpha_j Q_{jt} \right. \\ \left. + \sum_{j \in \mathcal{I}^{CHP}} \alpha_j (\rho_j^E P_{jt} + \rho_j^H Q_{jt}) \right), \end{aligned} \quad (35)$$

subject to DHN (1)-(4), temperature dynamics (7)-(13), nodal (14)-(20), HES (21)-(23), HS (24)-(30), and power system (31)-(33) constraints, over the set of optimization variables

$$\Omega = \{P, Q, L^{HS}, L^{pump}, mf^{HS}, mf^{HES}, mf^S, mf^R, T^S, T^R, T^{S,in}, T^{S,out}, T^{R,in}, T^{R,out}, \tilde{T}^{S,in}, \tilde{T}^{R,in}, \tau^S, \tau^R, u^S, v^S, u^R, v^R, B, \theta\}.$$

III. CONVEX RELAXATIONS

In the CHPD model presented above, non-convexities arise from the bilinear equality constraints (16), (17), (21), (24) and (27), and the quadratic equality constraints (2).

A. McCormick Envelopes

We linearize the bilinear terms in Eqs. (16)-(17), (21), (24), and (27) using a McCormick relaxation approach [15]. For each bilinear term we introduce a set of linear upper and lower bounding functions. The McCormick relaxation of a bilinear function $f : (x, y) \mapsto xy$ defines its convex envelope over the domain $[x^l, x^u] \times [y^l, y^u]$ [17]. The McCormick envelopes of the bilinear terms in (21) and (16) are presented in [18]. We do not detail this step for other constraints because the procedure is well-covered in the existing literature.

B. Convex Quadratic Programming

The pressure loss constraints in (2) can be made convex using a convex quadratic relaxation, a special case of SOCP [13], such that

$$\begin{aligned} pr_{n_1 t}^S - pr_{n_2 t}^S &\geq \nu_p (m_{pt}^S)^2, \quad pr_{n_2 t}^R - pr_{n_1 t}^R \geq \nu_p (m_{pt}^R)^2 \\ \forall n_1, n_2 \in \mathcal{I}^N, p \in S_{n_1}^{P+} \cap S_{n_2}^{P-}, t \in \mathcal{T}. \end{aligned} \quad (36)$$

As a result, the original MINLP is relaxed into an MISOCP and can be solved by commercial solvers. While MISOCP is a rapidly improving class of algorithms, current solvers do not scale to large problems. In order to address this issue, we next introduce an MILP relaxation.

C. Outer Approximation

The constraints representing pressure loss in pipes can be linearized using an outer approximation approach [19]–[21]. We introduce a set of feasible points $(pr_{n_1 t}^{S,l}, pr_{n_2 t}^{S,l})$ for $l = 1, \dots, L$ and use a Taylor's series expansion to linearize (2) around these points, such that

$$\begin{aligned} m_{pt}^S &\leq \frac{\sqrt{pr_{n_1 t}^{S,l} - pr_{n_2 t}^{S,l}}}{2\sqrt{\mu_p}} + \frac{(pr_{n_1 t}^S - pr_{n_2 t}^S)}{2\sqrt{\mu_p} (pr_{n_1 t}^{S,l} - pr_{n_2 t}^{S,l})} \\ \forall n_1, n_2 \in \mathcal{I}^N, p \in S_{n_1}^{P+} \cap S_{n_2}^{P-}, t \in \mathcal{T}, l \in \{1, \dots, L\}. \end{aligned} \quad (37)$$

We similarly use an outer approximation for the constraints representing pressure loss in the return network. As a result the original CHPD model is relaxed into an MILP and can be solved efficiently with commercial solvers.

D. Deriving Feasible Solutions

The solutions to the MISOCP and MILP relaxations introduced in Section III are not always feasible for the original MINLP model in Section II. Indeed the non convex constraints (16)-(17),(2),(21),(24),(27) might be violated. When this happens, in order to derive feasible solutions, we fix the binary variables related to time delays and mass flow rates and use an NLP solver to find a local optimal solution of the original problem.

There is no known method to analytically characterize the gap between the solution of the relaxed problem and the global optimum of the original MINLP. Comparing the solutions of the relaxed problem and the derived feasible solution of the original MINLP provides a bound on this gap. However this numerical analysis is extremely case-dependent.

IV. CASE STUDY

A. Case Study Setup

The proposed CHPD model is implemented on an integrated six-bus electricity and three-node district heating network described in Fig. 2. This system comprises a conventional

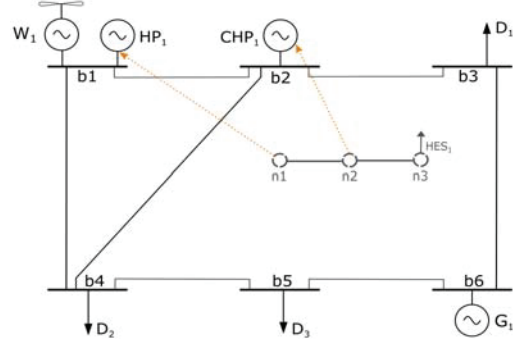


Figure 2. Integrated heat and electricity system

electricity generator G_1 , a wind producer W_1 , a HP and an extraction CHP. The technical characteristics of the networks and generation units are derived from [10], and detailed in [18]. The CHPD is tested over 24 hourly time steps. We use wind production data from [22]. Heat demand peak hours occur during the evening and morning, due to lower ambient temperatures, while electricity demand peak hours are during the day [23]. Heat and electricity loads and available wind production are represented in Fig. 3.

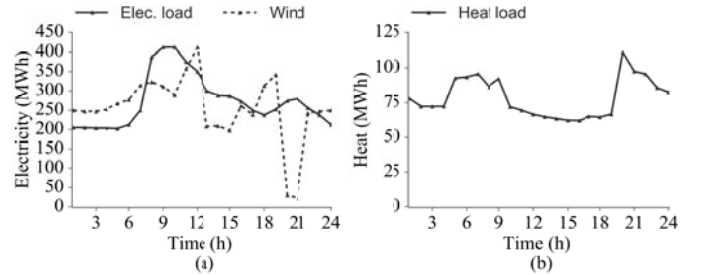


Figure 3. Case study setup: (a) Electricity load and available wind power (b) heat load

In order to assess the impact of the DHN on the power system, the proposed CHPD is compared to a Conventional Economic Dispatch (CED). The CED is a co-optimization model for heat and electricity, in which the DHN is not modeled. The equations representing DHN (1)-(13), nodal (14)-(20), HS (24),(26)-(27) and HES (21)-(23) constraints are replaced by the following heat balance equations

$$\sum_{i \in \mathcal{I}^{HES}} L_{it}^H = \sum_{j \in \mathcal{I}^{HS}} Q_{jt} \quad \forall t \in \mathcal{T}. \quad (38)$$

Since water pumps located at heat stations are not modeled, the power balance equation (31) is replaced by

$$L_{nt}^E + \sum_{j \in S_n^{HP}} L_{jt}^{HP} = \sum_{j \in S_n^E} P_{jt} + \sum_{m \in S_n^B} B_{nm} (\theta_{mt} - \theta_{nt}), \quad (39)$$

and other power system constraints (32)-(33) remain unchanged. This optimization model is detailed in [18]. Both CHPD and CED models are implemented on the integrated heat and electricity system presented above. The solutions of the CHPD are obtained by solving the MILP relaxation in Section III. When these solutions are infeasible for the original MINLP, we derive feasible solutions as discussed in Section III-D.

B. Results

Fig. 4 represents the electricity dispatch with the CHPD and CED models, and Fig. 5 the heat dispatch. In order to comparable the results for both models, we did not include electricity consumption from water pumps in this figure. With

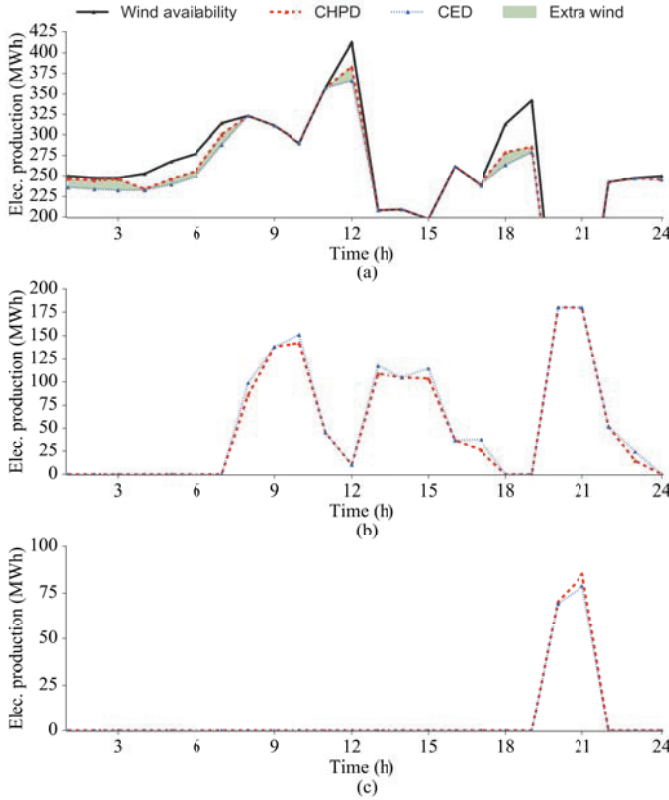


Figure 4. Electricity dispatch of (a) wind producer (b) conventional generator and (c) CHP

the CED model, the heat production from the HP and CHP balances the demand at each time period. In the CHPD model, the DHN provides operational flexibility to the energy system by temporally decoupling heat production and consumption (see Fig. 6(a)). This allows the HP to increase its production in the hours preceding heat peak load and when there is a wind surplus. As a result, an increase in wind utilization during these

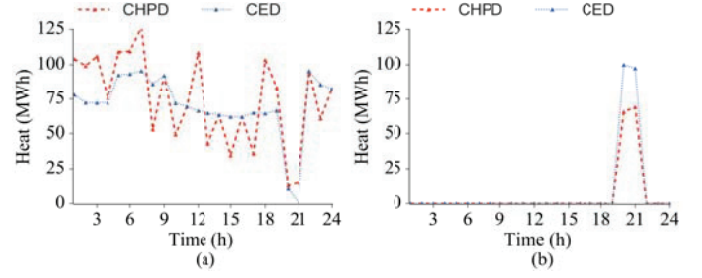


Figure 5. Heat dispatch of (a) HP and (b) CHP

hours can be observed in Fig. ???. Although wind curtailment occurs in both models during electricity off-peak hours, and especially at night, the CHPD increases the wind utilization by 100.8MWh, or 1.65%.

As shown in Fig. 6, the DHN acts as an energy storage. During charging periods, the energy injected in the pipes is

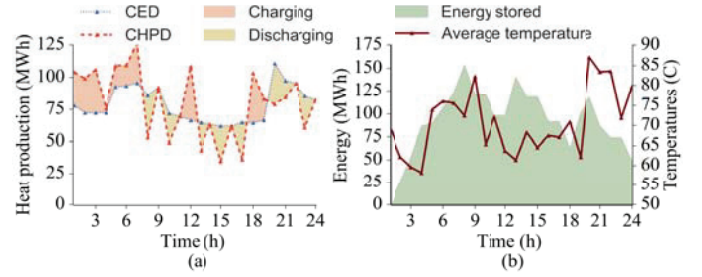


Figure 6. Storage capacity of the DHN: (a) heat production and consumption (b) average temperatures and total energy stored in pipelines

higher than the energy consumed. During discharging periods, the heat system consumes more energy than what is produced. We exploit this flexibility by controlling both inlet temperatures and mass flow rates in the pipelines. Charging periods correspond to high wind production hours, and discharging periods to high heat demand hours.

Fig. ?? shows that the conventional generator is used to cover electricity peak loads. With the CHPD, we observe a decrease in its utilization during discharging periods, thanks to the flexible operation of the HP. And during periods with low wind production, the CHP is dispatched to cover both heat and electricity demands, as seen in Fig. ??.

The production costs of the CHPD and CED models are respectively 18600\$ and 19277\$. The CHPD improves the operating cost of the system by 3.51%. This simulation shows that the CHPD model improves the economic efficiency of the overall energy system and increases wind utilization by exploiting the flexibility of the DHN.

Finally, the dispatch obtained with the CED model is not guaranteed to be feasible in practice because it neglects temperature dynamics in the pipes. An additional hydraulic check and, when necessary, a redispatch should be performed.

V. CONCLUSION

The CHPD introduced in this paper models the temperature dynamics in pipelines and accounts for the energy storage

capacity of DHNs. By allowing both temperatures and mass flow rates to vary, this model exploits the flexibility of the DHN. In order to solve the resulting MINLP, two convex relaxations are proposed. The small case study presented in Section IV shows the benefits of the CHPD model compared to a conventional economic dispatch, in terms of wind power utilization and production costs.

This work opens up several options for market design in support of renewable integration. First, in order to deal with the increased uncertainty from renewable energy sources, optimization under uncertainty approaches should be investigated. In Ref. [24], co-optimization of energy and reserves is proposed, via robust optimization and linear decision rules. This general framework could be applied to a CHPD model in order to link the uncertainties from the electricity system with the flexibility from the DHN, and co-optimize energy and reserves. Furthermore, a CHPD is not directly applicable in the current energy market framework. Indeed heat and electricity markets are still operated sequentially [6]. However, decentralized optimization approaches for markets coordination, such as the Alternating Direction of Multipliers Method (ADMM), could be investigated [25], [26].

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REFERENCES

- [1] P. Pinson, L. Mitridati, C. Ordoudis, and J. Ostergaard, "Towards fully renewable energy systems: Experience and trends in Denmark," *J. Power Energy Syst.*, vol. 3, no. 1, pp. 26–35, 2017.
- [2] H. Lund, B. Möller, B. V. Mathiesen, and A. Dyrelund, "The role of district heating in future renewable energy systems," *Energy*, vol. 35, no. 3, pp. 1381–1390, 2010.
- [3] P. Meibom, K. B. Hilger, H. Madsen, and D. Vinther, "Energy comes together in Denmark: The key to a future fossil-free Danish power system," *IEEE Power Energy Mag.*, vol. 11, no. 5, pp. 46–55, 2013.
- [4] R. Lahdelma and H. Hakonen, "An efficient linear programming algorithm for combined heat and power production," *Eur. J. Oper. Res.*, vol. 148, no. 1, pp. 141–151, 2003.
- [5] A. Rong, H. Hakonen, and R. Lahdelma, "A dynamic regrouping based sequential dynamic programming algorithm for unit commitment of combined heat and power systems," *Energy Convers. Manage.*, vol. 50, no. 4, pp. 1108–1115, 2009.
- [6] M. Zugno, J. M. Morales, and H. Madsen, "Commitment and dispatch of heat and power units via finely adjustable robust optimization," *Comput. & Oper. Res.*, vol. 75, pp. 191–201, 2016.
- [7] B. C. Ummels, M. Gibescu, E. Pelgrum, W. L. Kling, and A. J. Brand, "Impacts of wind power on thermal generation unit commitment and dispatch," *IEEE Trans. Energy Convers.*, vol. 22, no. 1, pp. 44–51, 2007.
- [8] X. Chen, C. Kang, M. O'Malley, Q. Xia, J. Bai, C. Liu, R. Sun, W. Wang, and H. Li, "Increasing the flexibility of combined heat and power for wind power integration in China: Modeling and implications," *IEEE Trans. Power Syst.*, vol. 30, no. 4, pp. 1848–1857, 2015.
- [9] Z. Li, W. Wu, J. Wang, B. Zhang, and T. Zheng, "Transmission-constrained unit commitment considering combined electricity and district heating networks," *IEEE Trans. Sustain. Energy*, vol. 7, no. 2, pp. 480–492, 2016.
- [10] Z. Li, W. Wu, M. Shahidehpour, J. Wang, and B. Zhang, "Combined heat and power dispatch considering pipeline energy storage of district heating network," *IEEE Trans. Sustain. Energy*, vol. 7, no. 1, pp. 12–22, 2016.
- [11] S. H. Low, "Convex relaxation of optimal power flowpart i: Formulations and equivalence," *IEEE Trans. Control Network Syst.*, vol. 1, no. 1, pp. 15–27, 2014.
- [12] —, "Convex relaxation of optimal power flowpart ii: Exactness," *IEEE Trans. Control Network Syst.*, vol. 1, no. 2, pp. 177–189, 2014.
- [13] H. Hijazi, C. Coffrin, and P. Van Hentenryck, "Convex quadratic relaxations for mixed-integer nonlinear programs in power systems," *Math. Programming Comput.*, vol. 9, no. 3, pp. 321–367, 2017.
- [14] C. Borraz-Sánchez, R. Bent, S. Backhaus, H. Hijazi, and P. V. Hentenryck, "Convex relaxations for gas expansion planning," *INFORMS J. Comput.*, vol. 28, no. 4, pp. 645–656, 2016.
- [15] G. P. McCormick, "Computability of global solutions to factorable nonconvex programs: Part iconvex underestimating problems," *Math. Programming*, vol. 10, no. 1, pp. 147–175, 1976.
- [16] G. Sandou, S. Font, S. Tebbani, A. Hirt, and C. Mondon, "Predictive control of a complex district heating network," in *Conf. Decision Control*, vol. 44, no. 8. IEEE, 2005, p. 7372.
- [17] J. Luedtke, M. Namazifar, and J. Linderoth, "Some results on the strength of relaxations of multilinear functions," *Math. Programming*, vol. 136, no. 2, pp. 325–351, 2012.
- [18] L. Mitridati and J. Taylor, "Power systems flexibility from district heating networks: Online appendix," March. 2018. [Online]. Available: <https://doi.org/10.5281/zenodo.1195508>
- [19] M. A. Duran and I. E. Grossmann, "An outer-approximation algorithm for a class of mixed-integer nonlinear programs," *Math. Programming*, vol. 36, no. 3, pp. 307–339, 1986.
- [20] J. Viswanathan and I. E. Grossmann, "A combined penalty function and outer-approximation method for MINLP optimization," *Comput. & Chemical Eng.*, vol. 14, no. 7, pp. 769–782, 1990.
- [21] R. Fletcher and S. Leyffer, "Solving mixed integer nonlinear programs by outer approximation," *Math. Programming*, vol. 66, no. 1, pp. 327–349, 1994.
- [22] W. Bukhsh, "Data for stochastic multiperiod optimal power flow problem." [Online]. Available: <http://sites.google.com/site/datasmpof/>
- [23] H. Madsen, "Time series analysis. course notes." [Online]. Available: <http://www.imm.dtu.dk/~hmad/time.series.analysis/assignments/index.html>
- [24] J. Warrington, P. Goulart, S. Mariéthoz, and M. Morari, "Policy-based reserves for power systems," *IEEE Trans. Power Syst.*, vol. 28, no. 4, pp. 4427–4437, 2013.
- [25] C. He, L. Wu, T. Liu, and M. Shahidehpour, "Robust co-optimization scheduling of electricity and natural gas systems via admm," *IEEE Trans. Sustain. Energy*, vol. 8, no. 2, pp. 658–670, 2017.
- [26] C. Wang, W. Wei, J. Wang, L. Bai, and Y. Liang, "Distributed optimal gas-power flow using convex optimization and admm," *arXiv preprint arXiv:1610.04681*, 2016.